Lodz University of Technology
Faculty of Technical Physics, Information Technology and Applied Mathematics
Institute of Information Technology

Selected Issues of Network Applications
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Modern Service Mesh - a better way to secure communication between microservices.

Jędrzej Bartoszewski, 251482
Michał Domagała, 251505
Kacper Maziarz, 251586
Mikołaj Pawłoś, 258681

1. Problem description

I. *Definition of the problem*

With the modern shift to microservice and containerized architectures, the industry decided to change course with its tooling, migrating from simple VMs to container orchestrators such as Kubernetes. Yet, in terms of securing the channels of transfer, we haven't seen such a breakthrough. We still use manual, laborious ways of securing inter-application communication, which is more prone to human error, creates a bigger opportunity for vulnerabilities, and increases time-to-market (which is now more important than ever).

II. *Description and justification of the proposed solution*

We propose implementing a Zero Trust Architecture using a modern service mesh. We will deploy a simulated microservice environment where every single network request between internal services is intercepted by an ultra-lightweight "sidecar" proxy. This proxy will enforce mutual TLS (mTLS) encryption and cryptographic identity verification for every connection. This project will contribute to the state-of-the-art by demonstrating how cutting-edge service meshes, such as Linkerd, provide significant gains by decoupling and centralizing security, routing, and observability. This takes the responsibility away from the application code; a benefit the industry hasn't fully realized yet, perhaps discouraged by the complexity of older service mesh implementations.

III. *Existing alternatives*

Existing approaches to securing communication between microservices include application-level security, manually configured TLS/mTLS, and network-level security mechanisms such as Kubernetes Network Policies. While these solutions can provide effective protection, they often require additional configuration, changes to application code, or ongoing manual maintenance. As a result, security policies may become inconsistent across services, increasing operational complexity and the risk of human error.

Service meshes such as Istio and Linkerd offer an alternative by centralizing security enforcement and automating features such as mTLS, service identity verification, and traffic management. Istio provides extensive configuration options and advanced capabilities, but its complexity and resource requirements may introduce additional operational overhead. Linkerd focuses on simplicity and lightweight operation, making it a suitable choice for demonstrating Zero Trust principles in a containerized environment.

2. Selected tools

- **k3d:** A lightweight Kubernetes distribution that runs Kubernetes clusters inside Docker containers. It allows setting up a cluster inside single-node environments or on a local machine (e.g., PC). Also removes the need of using paid cloud k8s engines (e.g., AKS EKS, GKS).

- **Linkerd**: An ultra-lightweight Service Mesh that provides automatic mTLS encryption, service identity verification, and granular Zero-Trust access control policies without requiring application code changes.

- **Grafana**: A visualization dashboard platform providing real-time insights into collected metrics. Charts, graphs, and visual metrics will highlight the new functionalities provided by the Service Mesh.

- **Prometheus / VictoriaMetrics**: Time-series databases collecting network metrics and policy audit logs from Service Mesh services. They provide the data needed to audit system health and verify policy enforcement during attack simulations.

- **Example microservice application (e. g. Google Online Boutique)**: A distributed application serving as a testing environment for the proposed solution.

3. Functional requirements

The project will provide a simulated microservice environment deployed on Kubernetes, secured using a service mesh implementing Zero Trust principles. The system will include the following functionalities:

- **Automated mTLS encryption:** Automatically encrypt communication between microservices using mutual TLS, without requiring changes to application code.
- **Service identity verification:** Authenticate services using cryptographic identities and prevent unauthorized services from establishing trusted connections.
- **Access control:** Configure and enforce policies defining which services are permitted to communicate with one another.
- **Observability:** Collect and present relevant metrics and communication data to monitor service interactions and security-related behavior.
- **Security validation:** Test the environment under different configurations, including unsecured communication, enforced mTLS, and restricted access policies, to verify the effectiveness of the implemented mechanisms.

4. Schedule

1) Set up the k3d cluster and deploy the example microservice application (e.g. Google Online Boutique) [Week 3-4]
2) Deploy a Service Mesh solution (Linkerd) with a basic configuration to demonstrate encrypted service-to-service communication (mTLS) and basic traffic verification as a Proof of Concept [Week 5-6] – **Deadline: 6.11.2026**
3) Design and implement strict Zero-Trust access policies in Linkerd [Week 7]
4) Integrate Grafana and Prometheus with Linkerd for continuous security and traffic monitoring [Week 8]
5) Prepare and run simple attack simulation on the application infrastructure to present increased security and visibility of the microservice network layer [Week 9] – **Deadline: 27.11.2026**

5. Work division

Member	**Responsibility**
Mikołaj Pawłoś	Set up the Kubernetes (k3d) cluster and deploying the example microservice application.
Jędrzej Bartoszewski	Integrating a Service Mesh solution (Linkerd) into the app environment, basic configuration for mTLS and POC purposes
Michał Domagała	Designing and launching the attacks on the enhanced version of the system's environment as a live-proof, implementing Zero-Trust access policies in Service Mesh
Kacper Maziarz	Integrate visualising, log- and metrics-aggregating tools with the Service Mesh solution